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Scaling in energy dependence of relative particle production rates and peak position of ξ distributions in e^+e^- annihilations¹

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Abstract

With the production rates, $\langle n_{\text{part}} \rangle$, of the K^0 , K^+ , p and Λ and average charged particle multiplicities, $\langle n_{\text{ch}} \rangle$, recently measured at LEP1.5–LEP2, scaling of the relative particle production rates $\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$ is seen in a broad center of mass energy range from $\sqrt{s} = 29\text{--}35$ to 183 GeV. The quantity $\xi_0 / \langle n_{\text{ch}} \rangle^{1/2}$, where ξ_0 is the position of the maximum of the $\xi = -\ln(2p/\sqrt{s})$ distribution and p is the particle momentum, also scales from 14 to 183 GeV. It is shown that the gaussian shape of the ξ distribution is a simple consequence of longitudinal phase space. The slope parameter in the logarithmic increase of ξ_0 with the center of mass energy must be smaller than 1, as observed experimentally, but this does not necessary imply a manifestation of coherence. © 1998 Published by Elsevier Science B.V. All rights reserved.

1. Introduction

The precise LEP1 data allow not only stringent tests of electroweak predictions of the Standard Model but also many tests of QCD and detailed studies of hadron production which shed a new light on the production properties of hadrons in e^+e^- annihilations and even in hadronic interactions.

At LEP2, QCD tests and studies of hadron production properties are more difficult due to important initial state radiation, the WW production back-

ground, and limited statistics. Still, it is challenging to test the regularities observed in hadron production at the Z^0 peak in a new energy domain. The LEP experiments already measured the production rates of the K^0 , K^+ , p and Λ , the average charged particle multiplicities, $\langle n_{\text{ch}} \rangle$, the $\xi = -\ln(2p/\sqrt{s})$ spectra of particles where p is the particle momentum, and the evolution with the center of mass energy of the peak position, ξ_0 , of the ξ distribution in the reaction $e^+e^- \rightarrow q\bar{q}$ in the new energy range [1–14].

In this paper we use the results obtained by the LEP experiments [1–25] to show that scaling in the relative particle production rates, $\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$, previously observed at PETRA/PEP–LEP1 energies [26], is now also seen at higher energies. Then we show that the variable $\xi_0 / \langle n_{\text{ch}} \rangle^{1/2}$ exhibits scaling behaviour from $\sqrt{s} = 10$ to 183 GeV. The results taken from [1,3,7,8,11,12,22] are preliminary. Fi-

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Table 1

Total K^+ , K^0 , p and Λ rates, $\langle n_{\text{part}} \rangle$, and average charged particle multiplicities, $\langle n_{\text{ch}} \rangle$, per hadronic Z^0 decay and for the reaction $e^+e^- \rightarrow q\bar{q}$ at 133, 161, 172 and 183 GeV measured by the LEP experiments (A = ALEPH, D = DELPHI, L = L3, O = OPAL) and corresponding values of $\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$

Energy (GeV)	Particle	$\langle n_{\text{part}} \rangle$	Experiment	$\langle n_{\text{ch}} \rangle$	Experiment	$\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$
Z^0 peak	K^+	2.21 ± 0.06	ADO	20.99 ± 0.14	ADLO	0.105 ± 0.003
	K^0	2.014 ± 0.024	ADLO			0.0960 ± 0.0013
	p	1.02 ± 0.03	ADO			0.0486 ± 0.0015
	Λ	0.372 ± 0.007	ADLO			0.0177 ± 0.0004
133	K^+	2.58 ± 0.16	D	23.97 ± 0.30	ADLO	0.108 ± 0.007
	K^0	2.51 ± 0.25	D			0.105 ± 0.011
	p	1.25 ± 0.09	D			0.052 ± 0.004
	Λ	0.50 ± 0.09	D			0.021 ± 0.004
161	K^+	2.34 ± 0.27	D	25.42 ± 0.40	ADLO	0.093 ± 0.011
	K^0	2.56 ± 0.45	D			0.102 ± 0.018
	p	1.13 ± 0.13	D			0.045 ± 0.005
172	K^+	2.95 ± 0.23	D	26.54 ± 0.38	ADL	0.111 ± 0.009
	p	1.41 ± 0.17	D			0.053 ± 0.007
183	K^+	2.87 ± 0.21	D	26.64 ± 0.31	ADL	0.108 ± 0.008
	K^0	2.09 ± 0.24	D			0.079 ± 0.009
	p	1.32 ± 0.09	D			0.050 ± 0.003
	Λ	0.42 ± 0.11	D			0.016 ± 0.004

nally, we discuss the properties of the ξ distributions.

2. Scaling of $\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$

As was shown in [27], the mass dependence of the production rates of the pseudoscalar and vector mesons and the octet and decuplet baryons in e^+e^- annihilations, summed over all isospin projections and weighted with the spin counting factor $1/(2J+1)$, can be described (apart from pions) by a simple exponential

$$\frac{\langle N \rangle}{2J+1} = ae^{-bM^2}, \quad (1)$$

with the slope b independent of energy. Surprisingly similar behaviour, with the same slope b , was also observed in pp collisions for particles not resulting from fragmentation of the incident proton [28], and even for heavy ion collisions [29]. In e^+e^- annihilations, the energy dependence of parameter a in Eq. (1) is essentially determined by the energy dependence of the average charged particle multiplicity, $\langle n_{\text{ch}} \rangle$, revealing the existence of approximate scal-

ing of the relative particle production rates, $\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$, at least in the PETRA/PEP - LEP1 energy range [26]. Recently, the first results on the K^0 , K^+ , p and Λ production rates from 133 to 183 GeV were reported by the DELPHI experiment [1]. This allows scaling of the relative production rates to be tested at much higher energies.

The production rates of the K^0 , K^+ , p and Λ in the reaction $e^+e^- \rightarrow q\bar{q}$ at $\sqrt{s} = 133, 161, 172$ and 183 GeV measured by DELPHI [1] are given in Table 1. The values of $\langle n_{\text{ch}} \rangle$ obtained by averaging the results³ of the LEP experiments at 133 [3,4,9,13], 161 [3,6,10,14], 172 [3,6,10] and 183 GeV [3,7,12], together with the corresponding values of $\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$, are also collected in Table 1. The values of $\langle n_{\text{part}} \rangle$ at the Z^0 peak also given in Table

³ In calculating the errors on the averages, the procedure suggested by the PDG [30] was applied. If the quantity $\chi^2/(n-1)$, where $\chi^2 = \sum (\langle N \rangle - \langle N \rangle)^2 / (\Delta \langle N \rangle)^2$ is summed over the n experiments and $\langle N \rangle$ is the production rate, was larger than one, possibly indicating underestimated errors, the average error was multiplied by the scale factor $\sqrt{\chi^2/(n-1)}$.

1 have been obtained by averaging the production rates of the K^0 [17–20], K^+ and p [21–23], and Λ [17,19,24,25] measured by the LEP experiments. The corresponding $\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$ values at the Z^0 peak are obtained with $\langle n_{\text{ch}} \rangle$ taken from [30].

As seen from Table 1 and Fig. 1 where the values of $\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$ at $\sqrt{s} = 29\text{--}35$ GeV from PETRA/PEP are also presented (with the values of $\langle n_{\text{part}} \rangle$ and $\langle n_{\text{ch}} \rangle$ taken, respectively, from the PDG tables [30] and the compilation in [31]), the relative particle production rates scale with energy in the presently available energy range. Fits to the data by constants (horizontal lines in Fig. 1) give the values 0.0961 ± 0.0018 with $\chi^2/\text{ndf} = 8/4$ for the K^0 , 0.1059 ± 0.0023 with $\chi^2/\text{ndf} = 2/5$ for the K^+ , 0.0489 ± 0.0012 with $\chi^2/\text{ndf} = 2/5$ for the p and 0.0175 ± 0.0005 with $\chi^2/\text{ndf} = 6/3$ for the Λ . The value of χ^2/ndf for all four fits is 18/17. We verified that JETSET 7.4 and HERWIG 5.9 with the default parameters describe the energy independence

of $\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$ quite well, at least in the LEP1–LEP2 energy range.

Since the mass dependence of the production rates given by Eq. (1) is also valid for pp collisions with the same slope b as for e^+e^- annihilations [28] and since the energy dependence of $\langle n_{\text{ch}} \rangle$ in e^+e^- and $p^\pm p$ collisions is also very similar [32], the same scaling properties of $\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$ are expected for high energy $p^\pm p$ collisions, at least for particles not resulting from fragmentation of the incident hadrons. This is supported by the results obtained in the thermodynamical description of hadron production in $p^\pm p$ collisions [33], showing that the temperature, of about 170 MeV, of the hadron gas fireballs is the same as in e^+e^- collisions [34] and does not seem to depend on the center of mass energy. Therefore, the observation of scaling of $\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$, together with other regularities recently observed in particle production rates [27–29,32–34], strongly indicates that the production of at least light particles in high energy collisions is determined by a universal hadronization mechanism.

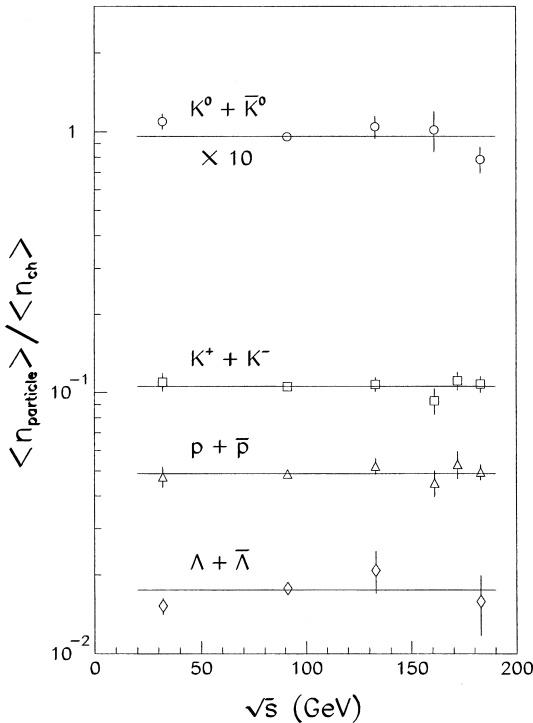


Fig. 1. The dependence of the relative particle production rates, $\langle n_{\text{part}} \rangle / \langle n_{\text{ch}} \rangle$, for the K^0 , K^+ , p and Λ on the center of mass energy. The lines are the fits to constant values whose results are given in the text.

3. Scaling of $\xi_0 / \langle n_{\text{ch}} \rangle^{1/2}$

The comparison of the energy evolution of other observables used to describe particle production in e^+e^- collisions with that of $\langle n_{\text{ch}} \rangle$ such as, for example, the position of the maximum, ξ_0 , of the ξ distribution might perhaps also be useful. Since $\langle n_{\text{ch}} \rangle$ is the integral of the ξ distribution, the properties of the ξ spectra suggest investigation of the energy variation of the variable $\xi_0 / \langle n_{\text{ch}} \rangle^{1/2}$. The values of ξ_0 and $\xi_0 / \langle n_{\text{ch}} \rangle^{1/2}$ for the different center of mass energies are collected in Table 2. The values of ξ_0 from 14 to 55 GeV were taken from [31], where the corresponding references to the experiments can be found. The values of $\langle n_{\text{ch}} \rangle$ from 14 to 55 GeV were also taken from [31] and those at 29, 35 and around 55 GeV measured by several experiments were averaged. For the LEP experiments, the values of $\langle n_{\text{ch}} \rangle$ given in Table 1 were used and the values of ξ_0 measured by the different experiments at the Z^0 peak [3,11,15,16] and higher energies [3,5,8,11–14] were averaged. As seen from Table 2 and Fig. 2, the variable $\xi_0 / \langle n_{\text{ch}} \rangle^{1/2}$ is practically independent of the center of mass energy in the presently available energy range. The fit of the data to a constant

Table 2

The values of ξ_0 and $\xi_0/\langle n_{\text{ch}} \rangle^{1/2}$ at the different center of mass energies

Energy (GeV)	ξ_0	Experiment	$\xi_0/\langle n_{\text{ch}} \rangle^{1/2}$
14	2.353 ± 0.043	TASSO	0.772 ± 0.022
22	2.651 ± 0.041	TASSO	0.789 ± 0.020
29	2.771 ± 0.015	MARK II	0.773 ± 0.009
29	2.886 ± 0.060	TPC	0.805 ± 0.019
34.8	3.063 ± 0.024	TASSO	0.831 ± 0.013
35	2.929 ± 0.072	CELLO	0.795 ± 0.022
43.6	3.120 ± 0.054	TASSO	0.803 ± 0.019
55	3.147 ± 0.093	AMY	0.776 ± 0.024
91.2	3.656 ± 0.018	ADLO	0.798 ± 0.005
133	3.887 ± 0.032	ADLO	0.794 ± 0.008
161	4.014 ± 0.035	ALO	0.796 ± 0.009
172	4.050 ± 0.045	AL	0.786 ± 0.010
183	4.100 ± 0.025	ADL	0.794 ± 0.007

(horizontal line in Fig. 2) gives 0.794 ± 0.003 with $\chi^2/\text{ndf} = 17/12$ (with the largest contribution of 11.4 to χ^2 given by the two data points of MARK II at $\sqrt{s} = 29$ GeV and TASSO at $\sqrt{s} = 35$ GeV). From this the following observations can be made.

Scaling of $\xi_0/\langle n_{\text{ch}} \rangle^{1/2}$ implies that the energy evolutions of the peak position, ξ_0 , and dispersion, σ , of the ξ distribution are essentially the same, as

was argued in [35]. The Modified Leading Logarithmic Approximation (MLLA) [36] under the assumption of Local Parton Hadron Duality (LPHD) [37] predicts

$$\xi_0 = Y \left[\frac{1}{2} + \sqrt{\frac{c}{Y}} - \frac{c}{Y} + O(Y^{-3/2}) \right] \quad (2)$$

and

$$\sigma \sim Y^{3/2}, \quad (3)$$

where $Y = \ln(\sqrt{s}/\Lambda_{\text{eff}})$, Λ_{eff} is the effective scale parameter and $c = 0.2915$ ($c = 0.3513$) for the active number of flavors $n_f = 3$ ($n_f = 5$). The ratio ξ_0/σ from (2) and (3) is also almost energy independent, increasing only by about 8% from 14 to 183 GeV for both values of n_f and for $\Lambda_{\text{eff}} = 150$ –250 MeV.

Scaling of $\xi_0/\langle n_{\text{ch}} \rangle^{1/2}$ implies that investigations of the energy dependence of $\langle n_{\text{ch}} \rangle$ and ξ_0 provide essentially the same information. Therefore if some model fails to describe the ξ distribution (like, for example, the COJETS Monte Carlo model [38] in the OPAL experiment [14]) or the energy evolution of ξ_0 , then it must also fail to describe the energy evolution of $\langle n_{\text{ch}} \rangle$ (as is the case for CO-

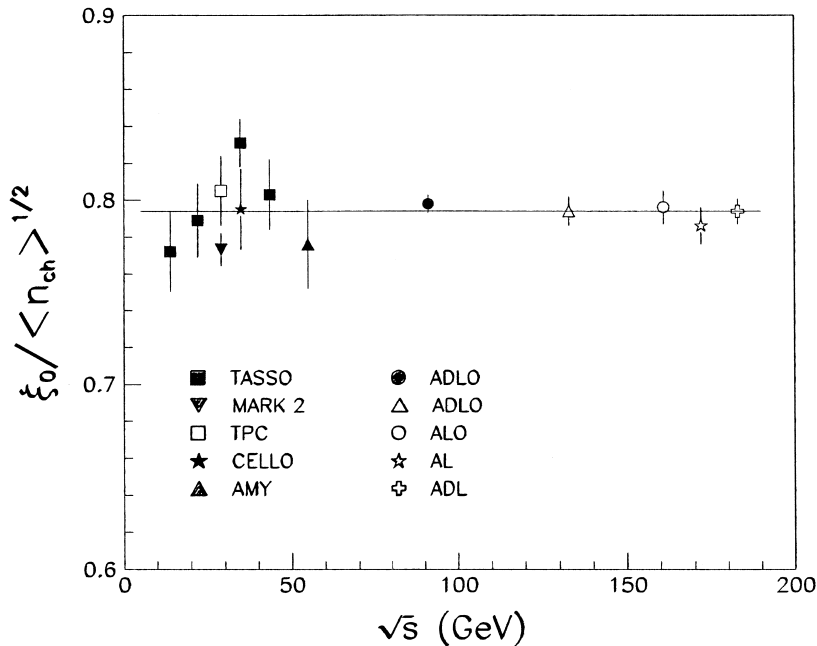


Fig. 2. The dependence of the variable $\xi_0/\langle n_{\text{ch}} \rangle^{1/2}$ on the center of mass energy; A = ALEPH, D = DELPHI, L = L3, O = OPAL. The line is the fit to a constant value whose result is given in the text.

JETS in [14]). In particular, if the energy evolution of ξ_0 indeed provides (or does not provide) evidence for the manifestation of coherence of soft gluons in the hadron momenta spectra, as argued in the MLLA approach under assumption of LPHD, then the same evidence must be presumably seen (or not seen) in the energy dependence of $\langle n_{ch} \rangle$.

4. Properties of the ξ spectra

The last observation allows us to return again to the question already raised in [35]: whether or not the gaussian shape of the ξ distribution or the observed logarithmic increase of the peak position ξ_0 with energy with a slope different from 1 can indeed be considered as a manifestation of color coherence of soft gluons in the hadron momentum spectra and, hence, as a proof of the LPHD hypothesis.

The exact relation between the differential cross section as a function of the variables ξ and θ , where θ is the production angle with respect to the thrust axis, or as a function of the longitudinal rapidity, y , and transverse mass, $m_T = (m^2 + p_T^2)^{1/2}$, is

$$\frac{d^2\sigma}{d\xi d\cos\theta} = \frac{2p^3}{E} \frac{d^2\sigma}{dy dm_T^2}. \quad (4)$$

The logarithmic dependence of ξ_0 on the center of mass energy and the gaussian shape of the ξ spectra is a trivial consequence of this transformation and of longitudinal phase space. Indeed, assuming factorization in y and m_T^2 and the usual parameterization of the transverse momentum distribution, one has:

$$\frac{d^2\sigma}{dy dm_T^2} = C(y) e^{-bm_T}. \quad (5)$$

Then, using the definition of $\xi = -\ln(2p/\sqrt{s})$ and neglecting particle masses for simplicity⁴, so that $y = \frac{1}{2} \ln \frac{1+\cos\theta}{1-\cos\theta}$, one gets

$$\frac{d^2\sigma}{d\xi d\cos\theta} = 2C(y) p^2 e^{-bp\sin\theta} = 2C(y) e^{f(\xi, \sin\theta)} \quad (6)$$

with

$$f(\xi, \sin\theta) = 2\ln \frac{\sqrt{s}}{2} - 2\xi - \frac{b\sqrt{s}\sin\theta}{2} e^{-\xi}. \quad (7)$$

From $f'_\xi = 0$, the maximum of the distribution (6) at fixed θ (i.e. at fixed rapidity y) is positioned at

$$\xi_0 = a + \ln\sqrt{s}, \quad (8)$$

with $a = \ln(\frac{1}{4}b\sin\theta)$.

It is important that the parameter a in Eq. (8) decreases with increasing center of mass energy, since the average transverse momentum increases with increasing energy and the parameter b in Eq. (5), related to the average transverse momentum, decreases with increasing energy. Therefore if the energy dependence of ξ_0 is parameterized as

$$\xi_0 = A + B\ln\sqrt{s}, \quad (9)$$

with A constant, the parameter B must be smaller than 1 by definition, due to the increase of the average transverse momentum with increasing energy.

The gaussian shape of the ξ distribution is also a simple consequence of Eq. (6). Indeed, with $\xi_0 = \ln(\frac{1}{4}b\sqrt{s}\sin\theta)$, the function (7) can be rewritten as

$$f(\xi, \sin\theta) = -2\ln \frac{b\sin\theta}{2} + 2(\xi_0 - \xi) - 2e^{\xi_0 - \xi}. \quad (10)$$

In the region of the peak, one can use the approximation

$$e^{\xi_0 - \xi} \approx 1 + (\xi_0 - \xi) + \frac{(\xi_0 - \xi)^2}{2}. \quad (11)$$

Then one obtains

$$\frac{d\sigma}{d\xi} = \text{const.} \int dy C(y) e^{-(\xi - \xi_0)^2}, \quad (12)$$

where

$$\xi_0 = \xi^* - \ln(\cosh y) \quad (13)$$

⁴ Numerical calculations without this assumption give the same results [35].

and $\xi^* = \ln(\frac{1}{4}b\sqrt{s})$ is the maximum of the ξ distribution at $y = 0$. Introducing a new variable $\lambda = \ln(\cosh y)$, Eq. (12) can be rewritten as

$$\frac{d\sigma}{d\xi} = \text{const.} \int_0^{\xi^*} d\lambda \left[C(y) \frac{\cosh y}{\sinh y} \right] e^{-(\xi - \xi^* + \lambda)^2}. \quad (14)$$

It is obvious from (14) that the position of the maximum of the ξ distribution depends not only on the value of the parameter b , as was shown earlier, but also on the shape of the rapidity spectrum. This can be illustrated by the following two simple examples. First, let us consider the flat rapidity spectrum, $C(y) = \text{const.}$ In this case, due to the increase of $\cosh y / \sinh y$ with decreasing y , the exponents with small values of λ give the main contribution to the integral (14), so the position of the maximum is very close to the value of ξ^* and the slope $B = 1$. In the second example, since the rapidity distributions in e^+e^- annihilations have a dip at $y \rightarrow 0$, we take $C(y) = \sinh y / \cosh y$ (the flat y spectrum at large y developing a large dip for $y \rightarrow 0$). In this case the contribution of all exponents in the integral (14) is the same, the maximum of the ξ distribution is positioned at $\xi^* / 2$ and the slope $B = 1/2$.

The experimental data (see, for example, [14] and references therein) follow the dependence (9), with a value of the slope parameter $B \approx 0.64$. The difference between this value and $B = 1$ is usually considered to be strong evidence for coherence. However, from the arguments presented above it is clear that the value of $B = 1$ represents the highest possible limit for the slope. In reality, even in absence of coherence, it must be smaller than one due to the increase of the average transverse momentum with the center of mass energy and the presence of the dip in the rapidity spectra at small y . The evolution of the rapidity spectra with the center of mass energy also influences the value of B . Therefore the difference between the experimentally measured value of B and $B = 1$ can not be considered as a definite proof of the manifestation of gluon coherence in the *hadron* momentum spectra and hence of the LPHD hypothesis.

The numerical integration of (12) over rapidity leads to a “hump-backed” shape of the ξ distribution with an approximately gaussian shape [35]. More

generally, the “hump-backed” shape of the ξ distribution follows *trivially and without any assumptions from Eq. (4)*. The fall-off of the spectrum with increasing momentum (decreasing ξ) at the left-hand side of the peak requires no comments. The fall-off at the right-hand side of the peak at small momenta (large ξ) is an *obvious property of the Jacobian* $2p^3/E$ in Eq. (4) for any reasonable shape of the rapidity distribution. This can be compared with the statements that “the observed structure of the spectra is dominated by physics of QCD-coherence, not being the artefact of one or another nonasymptotic effect (phase-space, charmed hadron decays and so on)” [39] or that “the emergence of the “hump-backed” plateau shape of the inclusive hadron spectrum in jets, recently definitely confirmed at LEP, is perhaps the most striking example of a highly non-trivial prediction of QCD where coherence plays a predominant role” [36]⁵. Similar statements can be also found in [40] and elsewhere.

The above arguments do not question the existence of soft gluon coherence in *parton* momentum spectra or the validity of the MLLA approach. The good agreement of the experimental ξ distributions and the energy evolution of the peak position with the MLLA predictions is rather impressive. However, since the “hump-backed” shape of the ξ distribution and even its approximately gaussian shape are dominantly determined by longitudinal phase space, it can not be considered as clear experimental evidence for coherence. Therefore, the possible manifestation of coherence in the properties of the ξ distributions can not be very different, at least from the experimental point of view, from its possible manifestation in the properties of other variables characterizing hadron production in e^+e^- collisions, such as, for example, in the rapidity spectra⁶ and especially in their dip at small y or in the energy evolution of the average charged particle multiplicity.

⁵ It is also quite amusing how often, after stating that the “hump-backed” shape of the ξ distribution is explained by coherence, one then argues that the energy evolution of the peak position in the absence of coherence (i.e. when according to the first statement, no peak is expected) should follow the logarithmic dependence with the slope $B = 1$.

⁶ This disagrees with the arguments presented in [39].

Since the experiments have no access to an alternative world without gluon coherence and its manifestation can be seen only by quantitative comparison of the experimental data with the theory, it is imperative for the theory to try to isolate the influence of coherence from all other effects in order to understand to which extent it may influence the *hadron momentum spectra*. If such an influence of coherence on the parton level is small, good agreement of the MLLA predictions with the data still provides very important information and, in particular, indicates the validity of the LPHD hypothesis. On the other hand, if the influence of coherence is large and the LPHD works, it should obviously manifest itself not only in the quantitative properties of the ξ distributions, but also in the properties of other observables.

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